SHORT SYLLABUS

BCSE419L Speech and Language Processing (3-0-0-3)

Levels of NLP - Morhpology: Derivational & Inflectional Morphology - POS tagging - Parsing - Semantics ; Corpora, Sentence Segmentation, Stemming- Word Embeddings ; Sentiment Classification, Named Entity Recognition, Text Summarization - Statistical and Deep Learning models ; Machine Translation - Encoder & Decoder Model, Attention Models, Question Answering ; Speech production – Perception of sound – Vocal tract model - Phonetics - Short-Time analysis of the signal – Energy – Zero crossing – Autocorrelation – Short time Fourier analysis; Feature represenation of speech :Mel Frequency Cepstral Coeffecients, PLP, LPCC, GFCC, i-vector ; Automatic Speech recognition formulation: Isolated word recognition – Large vocabulary continuous speech recognition - HMM/GMM based speech recognition – DNN/HMM model -- CNN based speech recognition - RNN language Models – Evaluation metrics, Speaker recognition model – Alexa/Google assistant based application development.

Course code	Course Title	L	T	P	C
BCSE419L	**Speech and Language Processing**	**3**	**0**	**0**	**3**
Pre-requisite	**NIL**	**Syllabus version**			
		1.0			

Course Objectives

1. Be competent with fundamental concepts for natural language processing and automatic speech recognition
2. To understand technologies involved in developing speech and language applications.
3. To demonstrate the use of deep learning for building applications in speech and natural language processing

Course Outcomes

At the end of this course, student will be able to:

1. Describe the importance of different NLP modules in Text processing and fundamentals of speech production
2. Describe ways to represent speech and text
3. Demonstrate the working of sequence models for text
4. Use signal processing techniques to analyze/represent the speech signal
5. Execute trials of speech/language systems

Module:1	**Introduction to Natural Language Processing**	**7 hours**
Overview of NLP - Introduction to Levels of NLP - Morhpology: Derivational & Inflectional Morphology - POS tagging - Parsing: Shallow and Dependency Parsing, Semantics: Word Level Semantics and Thematic roles.		
Module:2	**Text Preprocessing & Feature Representation**	**8 hours**
Introduction to Corpora, Sentence Segmentation, Stemming: Porter Stemmer, Bag of words and Vector Space Model, Topic Modeling, N-gram Language Model, Smoothing, Word Embeddings: Word2Vec, Glove and Fasttext.		
Module:3	**Applications of NLP-1**	**6 hours**
Sentiment Classification using ML & DL models, Named Entity Recognition - CRF and LSTMs, Text Summarization - Statistical and Deep Learning models.		
Module:4	**Applications of NLP-2**	**4 hours**
Machine Translation - Encoder & Decoder Model, Attention Models, Question Answering - Knowledge based Q&A and Deep Learning models for Q&A.		
Module:5	**Introduction to Speech Processing**	**6 hours**
Fundamentals of speech production – Perception of sound – Vocal tract model - Phonetics - Short-Time analysis of the signal – Energy – Zero crossing – Autocorrelation – Short time Fourier analysis.		
Module:6	**Feature Representaion of Speech Signal**	**4 hours**
Mel Frequency Cepstral Coeffecients, Perceptual linear prediction (PLP), Linear prediction cepstral coefficients (LPCC), Gammatone Frequency Cepstral Coefficients (GFCC), i-vector.		
Module:7	**Automatic Speech and Speaker Recognition**	**8 hours**
Automatic Speech recognition formulation: Isolated word recognition – Large vocabulary continuous speech recognition - HMM/GMM based speech recognition – DNN/HMM model -- CNN based speech recognition - RNN language Models – Evaluation metrics, Speaker recognition model – Alexa/Google assistant based application development.		
Module:8	Contemporary issues	**2 hours**
	Total Lecture hours:	**45 Hours**

Text Book(s)	
1.	Dan Jurafsky, James H. Martin “Speech and Language Processing”, Draft of 3rd Edition,Prentice Hall 2022.
2.	Jacob Benesty, M. M. Sondhi, Yiteng Huang "Springer Handbook of Speech Processing", Springer, 2008.
Reference Books	
1.	Uday Kamath, John Liu, James Whitaker "Deep Learning for NLP and Speech Recognition" Springer, ,2019.
2.	Steven Bird, Ewan Klein, Edward Loper "Natural Language Processing with Python", O'Reilly Media. 2009.
3.	Ben Gold, Nelson Morgan, Dan Ellis “Speech and Audio Signal Processing: Processing and Perception of Speech and Music”, John Wiley & Sons, 2011.
Mode of Evaluation: CAT / Written Assignment / Quiz / FAT	

Recommended by Board of Studies	09-05-2022		
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